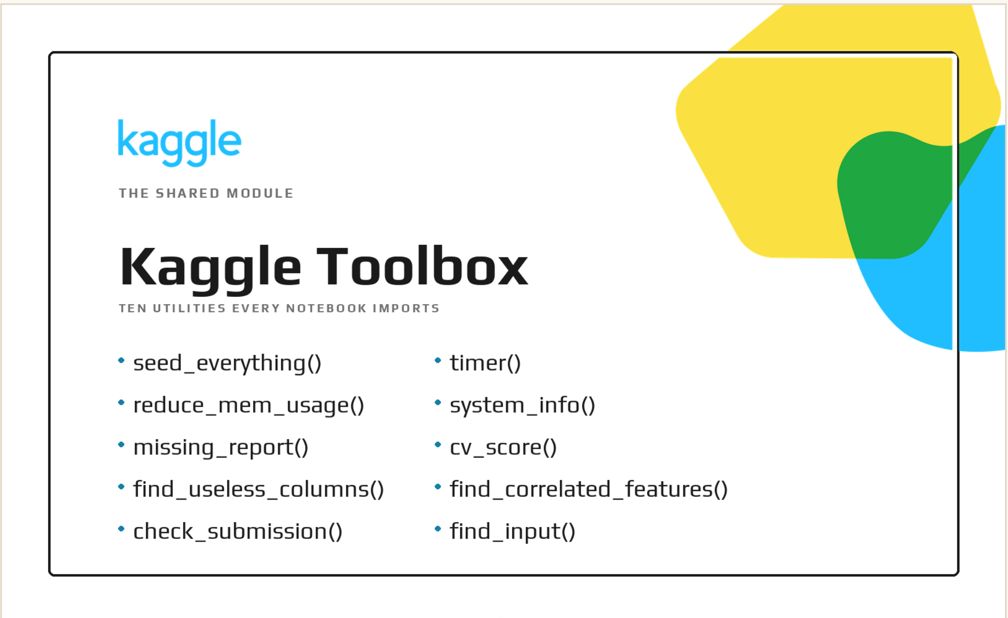
A WORKING METHOD

Competitive Data Science

The toolbox, the standard and the record behind a Kaggle profile

Notebooks Expert, ten bronze medals

10	17	50
TOOLBOX UTILITIES	COURSES	BADGES



Amey Thakur
github.com/Amey-Thakur
kaggle.com/ameythakur20

A Note from Amey Thakur

This repository holds the work behind a Kaggle profile: the notebooks entered into competitions, the reasoning that produced them, and the certificates and badges earned along the way. Each competition keeps its own folder, its own notebook and its own write-up explaining what the problem was, what was tried and where the approach ran out.

The write-ups are the point. A leaderboard score says how a submission placed and nothing about why, so each folder records the decisions: which signal the model was actually reading, which constraint shaped the design, and which of the three or four plausible approaches was taken and why.

What is in here

The toolbox	Ten utilities the notebooks share, each written because something went wrong without it.
The standard	The rules every notebook and write-up in the repository is held to, written out so the work stays consistent.
The method	One folder per entry, with the reasoning recorded whether or not the entry placed.
The record	The tier, the medals, the badges and the course certificates, kept with the work that earned them.

> **Note**
> This is an active repository. Entries are added as they are worked, and a folder is written when the work is done rather than when it succeeds. Some entries record approaches that did not place well and say so.

— Amey Thakur

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Start Here

What is in here.

10	17	50	10
TOOLBOX UTILITIES	COURSES	BADGES	BRONZE MEDALS

Who it is for

You are entering your first competition	The toolbox removes the failures everyone hits in week one, and the standard tells you what a finished entry looks like.
You already compete	A shared module and a written standard, offered as one way of keeping the work consistent across entries.
You are hiring or assessing	The reasoning is written down for every entry, including the ones that did not place.

> **Tip**
>

Four ways in

You want the utilities

Ten functions the notebooks share, each with the failure that caused it to be written.

Kaggle Toolbox

You want the method

The standard every notebook and write-up in the repository is held to.

The standard

You want to see the work

One folder per entry, with the notebook and the reasoning that produced it.

The repository

You want the profile

The notebooks, the medals and the tier, on Kaggle itself.

kaggle.com/ameythakur20

How to Use This Guide

Start with the toolbox	It is the part that transfers to your own work whatever you are entering.
Then the standard	It says what a notebook and a write-up here have to contain.
Entries are not listed here	The repository is active and the list changes. It lives in the repository, where it stays current.
Notebooks run on Kaggle	They depend on attached datasets and on competition data that cannot be redistributed.

> **Important**
> An independent record of my own work. Not affiliated with, sponsored by or endorsed by Kaggle or Google LLC. Kaggle is a trademark of Google LLC, used here only to identify the platform.

The Toolbox

The helper module every notebook imports. Each function is here because something went wrong without it.

Function	What it does	Why it exists
seed_everything(seed) Reproducibility	Locks Python random, NumPy, PyTorch on CPU and CUDA, TensorFlow and PYTHONHASHSEED, and enforces deterministic CUDA.	Uncontrolled randomness makes debugging unreliable and invalidates comparisons.
reduce_mem_usage(df) Memory optimisation	Downcasts each column to the smallest dtype its actual values allow.	Pandas defaults to int64 and float64, which is what crashes a kernel against a memory limit.
missing_report(df) Missing data analysis	Returns a sortable frame of missing count, missing percentage and dtype, for the columns that have any.	The standard output is neither sortable nor actionable.
find_useless_columns(df) Useless feature detection	Finds constant columns and duplicate columns.	They carry no signal, waste memory and add computation.
check_submission(sub, sample) Submission validation	Checks row count, column names and order, missing values, duplicate IDs and prediction range.	Submissions are limited, and an invalid file wastes an attempt.
timer(description) Code timing	A context manager that measures how long a block took.	Runtime limits mean you have to know where the time is going.
system_info() System diagnostics	Reports Python version, CPU cores, RAM, GPU and disk.	Environments vary, and the hardware changes both speed and reproducibility.
cv_score(model, X, y, cv, scoring) Cross-validation summary	Runs the fold loop and prints fold scores, mean, standard deviation and elapsed time.	Raw cross-validation output is hard to read and easy to misjudge.
find_correlated_features(df, t) Correlation filtering	Finds feature pairs above a correlation threshold and returns the ones worth removing.	Correlated features add redundancy and lengthen training.
find_input(pattern) Path finder	Searches the input, library and working directories and returns matching paths.	Input paths are dynamic, and hunting for them by hand wastes the first ten minutes of every notebook.

The Standard

Point an agent, or a person, at this folder and give it a competition, and it has everything needed to produce work that matches the rest of the repository.

File	What it fixes
AGENTS.md	The entry point: the prompt to paste, where files go, the badge colours and the fixed rules
NOTEBOOK-STANDARD.md	What a notebook in this repository must look like and must do
WRITEUP-STANDARD.md	What a write-up must record, so a reader learns the reasoning and not just the score

How Each Entry Is Written

The habits the repository is built on, which are the part worth copying.

Habit	What it means in practice
One folder per entry	The notebook and the write-up sit together, so the reasoning never drifts away from the code that produced it.
The write-up records decisions	Which signal the model was reading, which constraint shaped the design, and which of the plausible approaches was taken and why.
Failure is written down too	A folder is written when the work is done rather than when it succeeded. Approaches that did not place are recorded as such.
The toolbox is shared, not pasted	The same utilities are imported rather than copied into the top of every notebook.
Notebooks run where the data is	They depend on datasets attached in the Kaggle environment and on competition data that cannot be redistributed. Cloning gives you the code and the reasoning, not a runnable environment.

The Record

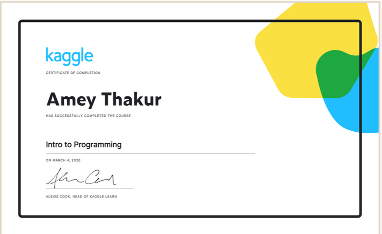
Kept with the work that earned it rather than listed on its own.

Notebooks Expert	The tier the profile currently holds
Ten bronze medals	Earned on notebooks, including the toolbox itself
Fifty badges	Every badge earned, each stored with its certificate
Seventeen courses	Kaggle Learn, from introductory programming through to computer vision, time series and machine learning explainability

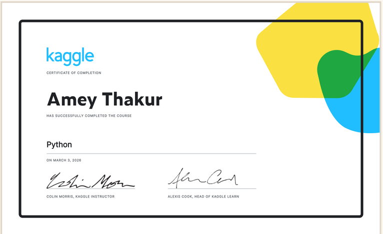
Note
Entries themselves are not listed in this document. The repository is active and the list changes; printing it here would date the page the week after it was written.

Kaggle Learn, Completed

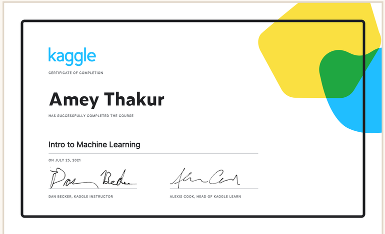
Seventeen courses, from introductory programming through to computer vision, time series and machine learning explainability. Every certificate is here, and every one opens.



Intro to Programming

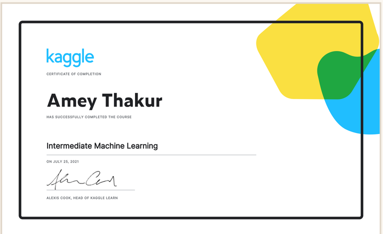


Python

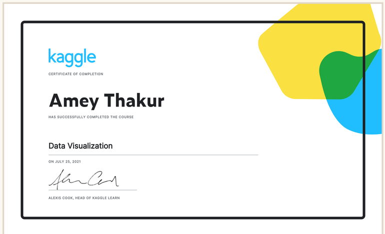


Intro to Machine Learning

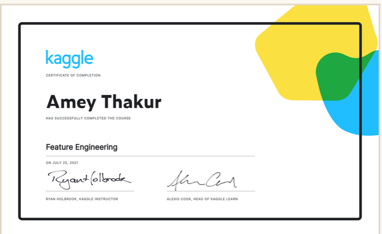
Pandas



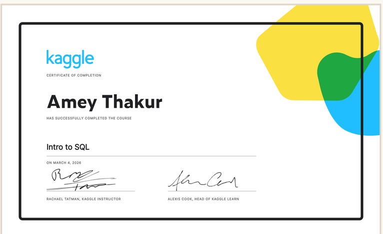
Intermediate Machine Learning



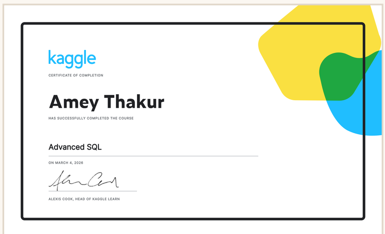
Data Visualization



Feature Engineering

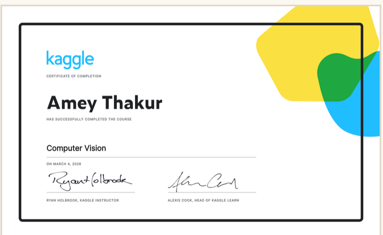


Intro to SQL

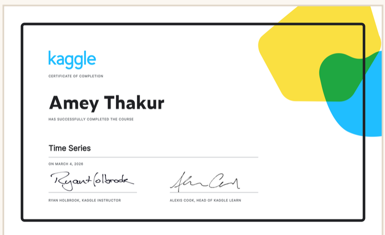


Advanced SQL

Intro to Deep Learning



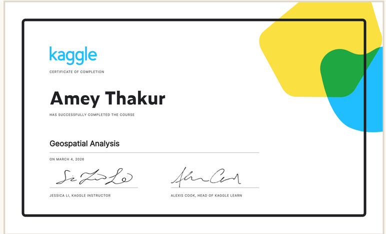
Computer Vision



Time Series

Data Cleaning

Intro to AI Ethics



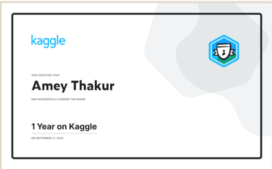
Geospatial Analysis

Machine Learning Explainability

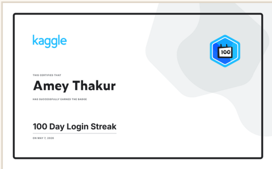
Intro to Game AI and Reinforcement Learning

Badges Earned

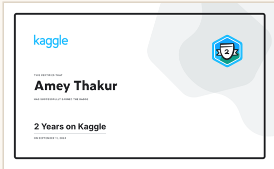
Fifty badges, each stored with the certificate that awarded it. They record what was actually done on the platform rather than what was entered.



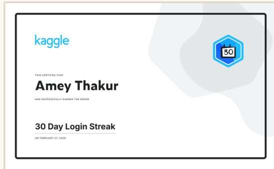
1 Year on Kaggle



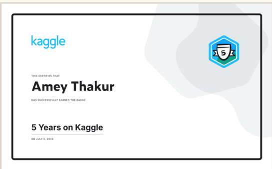
100 Day Login Streak



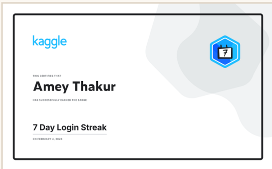
2 Years on Kaggle



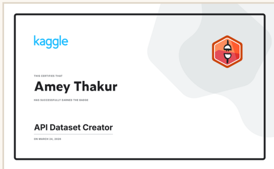
30 Day Login Streak



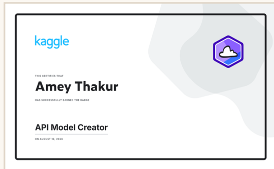
5 Years on Kaggle



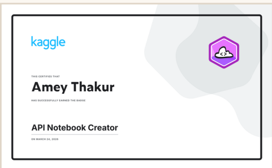
7 Day Login Streak



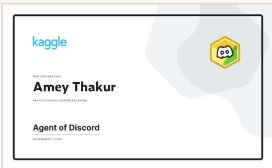
API Dataset Creator



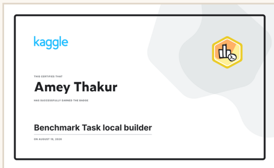
API Model Creator



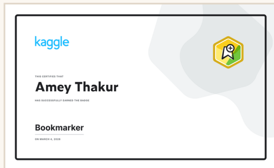
API Notebook Creator



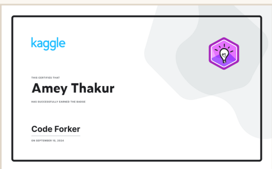
Agent of Discord



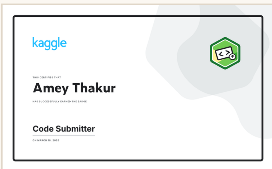
Benchmark Task local builder



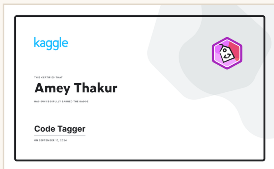
Bookmarker



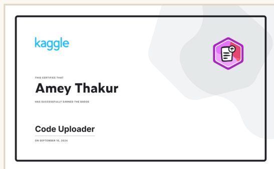
Code Forker



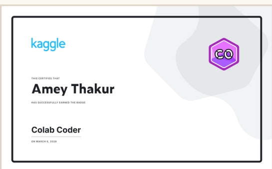
Code Submitter



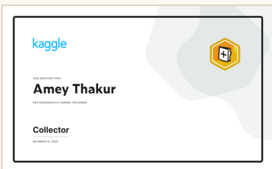
Code Tagger



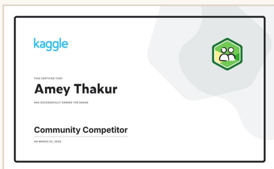
Code Uploader



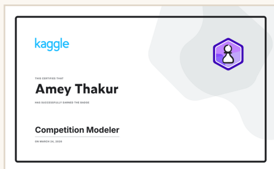
Colab Coder



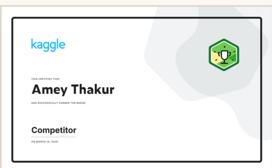
Collector



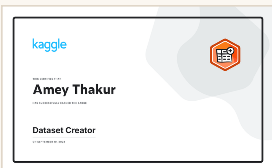
Community Competitor



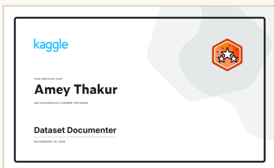
Competition Modeler



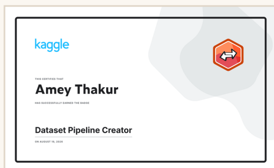
Competitor



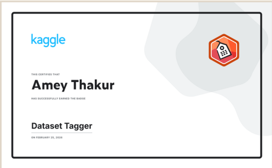
Dataset Creator



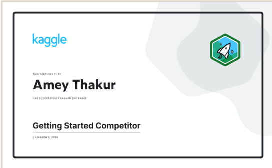
Dataset Documenter



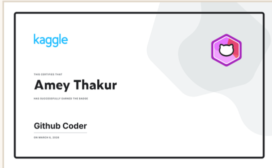
Dataset Pipeline Creator



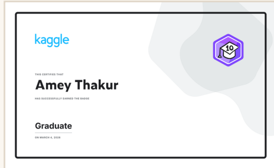
Dataset Tagger



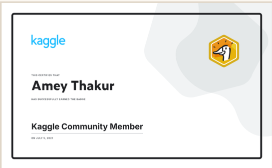
Getting Started Competitor



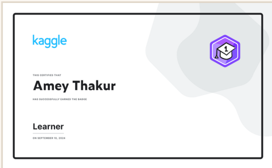
Github Coder



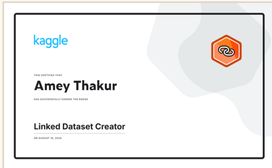
Graduate



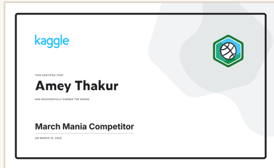
Kaggle Community Member



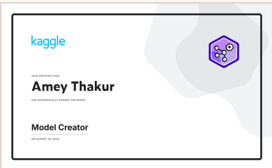
Learner



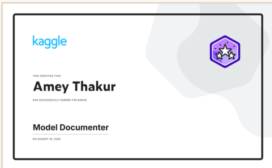
Linked Dataset Creator



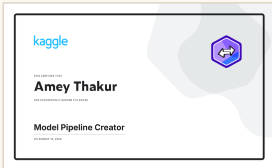
March Mania Competitor



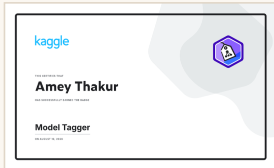
Model Creator



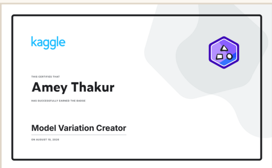
Model Documenter



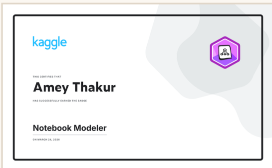
Model Pipeline Creator



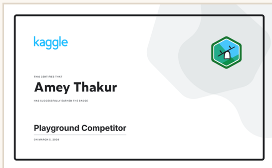
Model Tagger



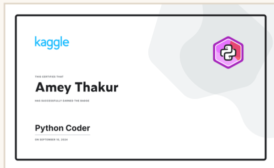
Model Variation Creator



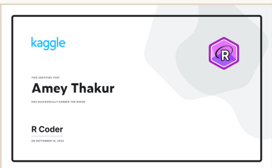
Notebook Modeler



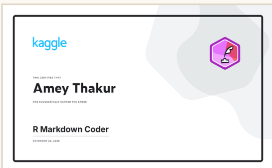
Playground Competitor



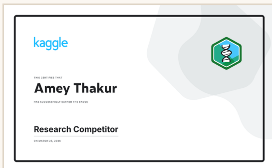
Python Coder



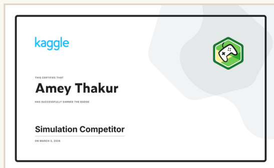
R Coder



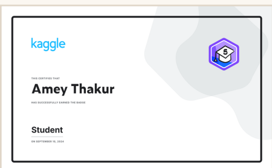
R Markdown Coder



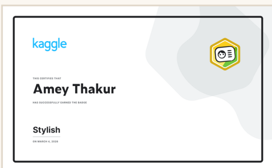
Research Competitor



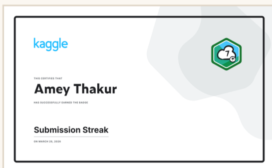
Simulation Competitor



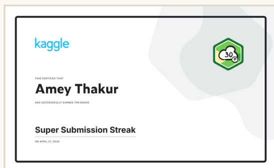
Student



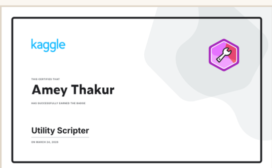
Stylish



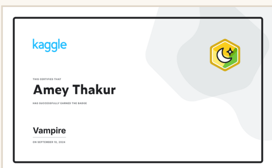
Submission Streak



Super Submission Streak



Utility Scripter



Vampire

What I Would Tell You

Six things I learned building this, offered for whatever they are worth.

1 **Write the reasoning down, not the score.**

A leaderboard position says how a submission placed and nothing about why. In a year the number is meaningless and the reasoning is the only thing left.

2 **Build the toolbox before you need it.**

Every function in mine exists because something went wrong without it. A submission validator costs an hour and saves a wasted attempt.

3 **Trust your local validation.**

The public leaderboard is a sample and it will lie to you. A cross-validation scheme that matches the split is worth more than a position.

4 **Seed everything, on day one.**

Without it you cannot tell whether a change helped or the random state moved. Every comparison you make after that is guesswork.

5 **Finish the entries that go badly.**

The write-up for an approach that did not work is the one you will actually reread. Publishing only the wins teaches you nothing.

6 **Do the free courses properly.**

Kaggle Learn is short and the certificates are worth little on their own. The habits they build are worth the days.

> **Tip**
> Competitions end and leaderboards reset. The method, the toolbox and the habit of writing down why are what carry over to the next one.

Explore Further

Where the rest of the work lives.

Related repositories

KAGGLE-COMPETITIONS	The repository this document indexes
Kaggle Toolbox	The shared module, as a notebook on Kaggle
Kaggle profile	The notebooks, medals and tier
RESEARCHGATE	Code and artefacts accompanying published work

Elsewhere

GitHub	github.com/Amey-Thakur
Amey's Arc	amey-thakur.github.io
ORCID	0000-0001-5644-1575
Google Scholar	Publications and citations
ResearchGate	researchgate.net/profile/Amey-Thakur
arXiv	arxiv.org/a/thakur_a_3
Kaggle	kaggle.com/ameythakur20
Hugging Face	huggingface.co/ameythakur
LinkedIn	linkedin.com/in/amey-thakur
YouTube	Engineering project demonstrations

A Closing Note

The repository is the work. This is the method behind it, written so that the part which transfers is the part you can read.

> **Important**
> Competitions close, datasets are withdrawn and leaderboards reset. The toolbox and the standard are what survive that, and they are the two things printed here.

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